3 **(a)** A solenoid shown in Figure 3.1 has length of 10 cm and 300 turns of wire. The current in the solenoid is 2.5 A when it is connected to a power supply.

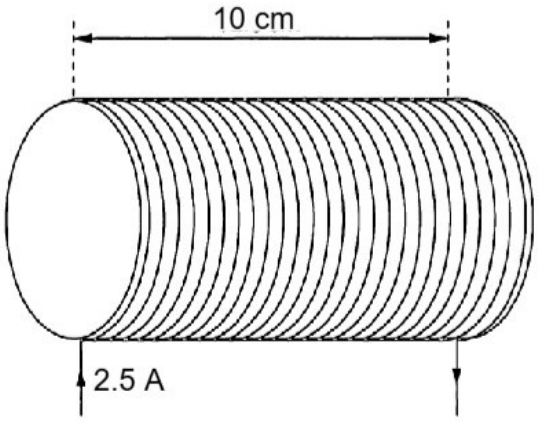


Figure 3.1

(i) Calculate the magnetic flux density inside the solenoid due to current of 2.5 A.

magnetic flux density = ...T [2]

(ii) Inside the dashed space provided in Figure 3.2, sketch the magnetic field pattern due to the current in the solenoid when 2.5 A of current is flowing in it as shown. Indicate the polarity of the magnetic field with **N** as the north pole and **S** as the south pole.

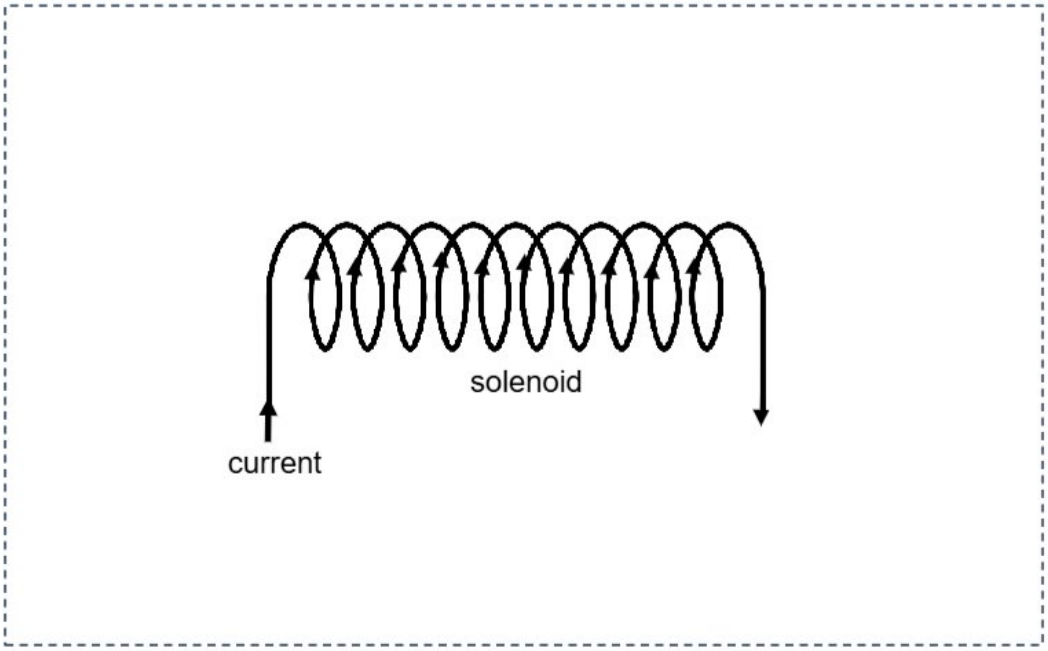


Figure 3.2 [2]

(b) A pair of concentric coils made of an inner coil Y placed within a solenoid is shown in Figure 3.3. The solenoid has cross-sectional area of 3.0×10^{-2} m^2 and is made of 2500 turns of wire. The solenoid is connected to a variable power supply by the terminals CD.

Coil Y has 500 turns of wire, a cross-sectional area of 7.5×10^{-3} m^2 and resistance of 3.0 Ω. Coil Y is connected to a resistor R of 15 Ω.

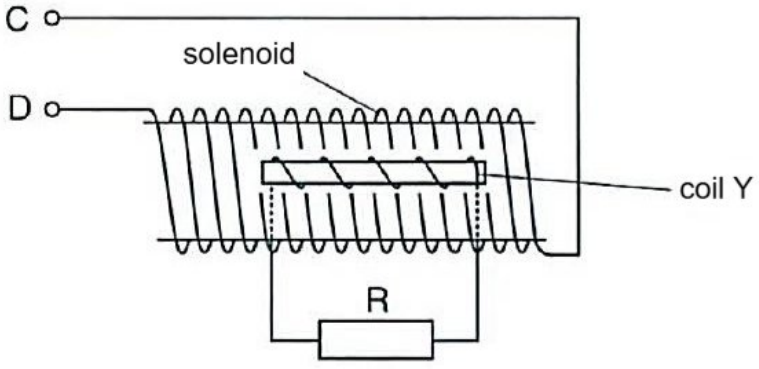


Figure 3.3

The current in the solenoid is then varied causing the magnetic flux density *B* in the solenoid to vary with time *t* as shown in Figure 3.4.

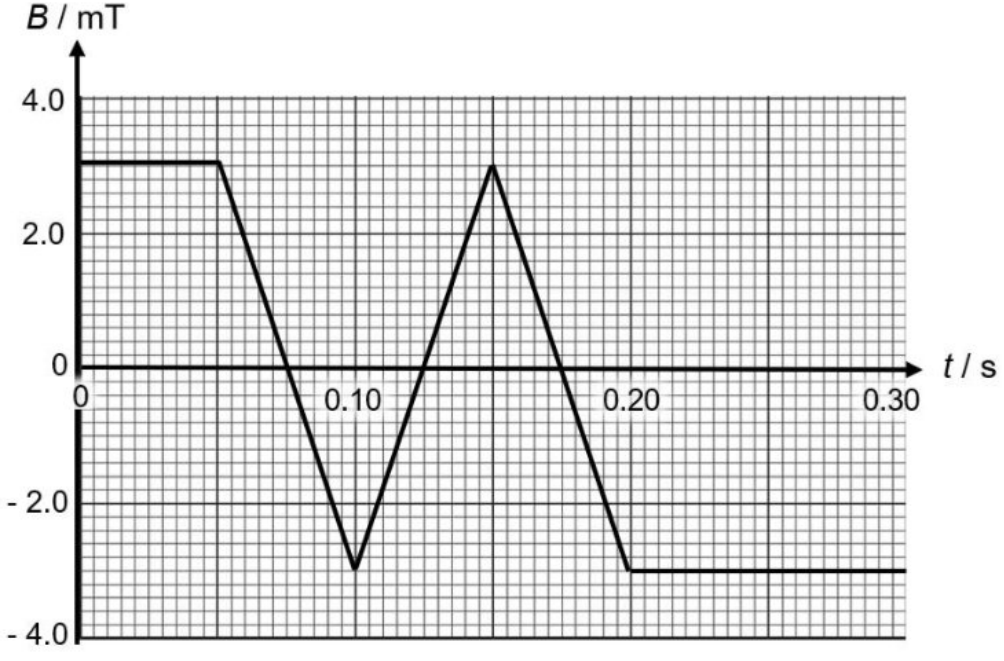


Figure 3.4

(i) Calculate the magnitude of the maximum current in R.

current = ...A [3]

(ii) On Figure 3.5, use your answer in **(b)(i)** to sketch a graph to show the variation with time of the current I in resistor R for time $t = 0$ to $t = 0.30$ s.

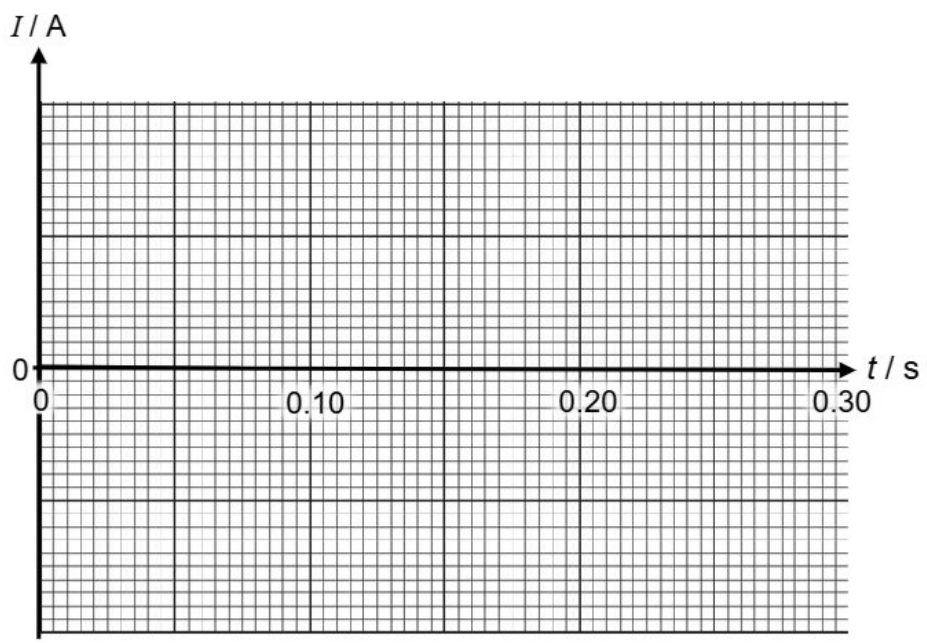


Figure 3.5

[2]

[Total: 9]